USN | | | | | | | | | | [23BSPH011]

SECOND SEMESTER B. Tech EXAMINATION, JUNE - 2024

ENGINEERING PHYSICS

Time: 3 Hours **Maximum Marks: 100**

Instructions:

I. Missing data may be suitably assumed.

II. Physical constants

- **i.** Mass of the electron $m_e = 9.1 \times 10^{-31}$kg
- **ii.** Mass of proton, neutron $m_n = m_p = 1.67 \times 10^{-27}$kg
- **iii.** Charge of the electron $e = 1.602 \times 10^{-19}$C
- **iv.** Planck's constant $h = 6.626 \times 10^{-34}$Js
- **v.** Boltzmann's constant $k = 1.38 \times 10^{-23}$J/K

ANSWER ALL QUESTIONS

		PART - A			**05x02=10**
1.	**a**	State De-Broglie`s hypothesis.	L2	CO1	**02**
	b.	What is Total internal Reflection?	L1	CO2	**02**
	c.	State fermi energy of an electron.	L2	CO3	**02**
	d.	What are shock waves?	L1	CO4	**02**
	e.	What are Elastic and Plastic bodies.	L2	CO5	**02**
		PART - B			**06x05=30**
2.	**a.**	Derive de-Broglie wavelength for an electron accelerated through a potential of V volts.	L2	CO1	**05**
		OR			
	b.	What are matter waves? and list their properties.	L2	CO1	**05**
3.	**a.**	What is a Wave function? Explain its physical significance.	L1	CO1	**05**
		OR			
	b.	Calculate the first 3 permitted energy values for an electron in a box of width 4 Å in eV.	L3	CO1	**05**
4.	**a.**	Explain the working principle of LIDAR.	L2	CO2	**05**
		OR			
	b.	With neat labelled diagram explain any two types of optical fibers.	L2	CO2	**05**
5.	**a.**	Review the assumptions of quantum free electron theory.	L3	CO3	**05**
		OR			
	b.	Explain the construction and working of a photo diode.	L2	CO4	**05**
6.	**a.**	Derive the spring constant for series combination of springs.	L2	CO5	**05**
		OR			
	b.	What are shock waves. Explain the properties of Shock Waves.	L2	CO5	**05**
7.	**a.**	Explain the different types of Modulus of Elasticity.	L2	CO6	**05**
		OR			

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	b.	What are the Theoretical and practical limits of Poisson`s ratio?	L3	CO6	05
		PART - C	**06x10=60**		
8.	a.	i. Describe Davisson and Germer`s experiment for confirmation of De-Broglie hypothesis.	L2	CO1	07
		ii. A particle of mass 0.5MeV $/c^2$ has kinetic energy 100eV. Find its de-Broglie wavelength, where c is the velocity of light	L3	CO1	03
		OR			
	b.	i. State Heisenberg`s uncertainty principle and Prove the Non-existence of an electron inside the Nucleus.	L3	CO1	06
		ii. An electron has a speed of 100m/s. The inherent uncertainty in its measurement is 0.005%. Calculate the corresponding uncertainty in finding its position.	L3	CO1	04
9.	a.	Set up time independent one-dimensional Schrodinger's wave equation.	L2	CO1	10
		OR			
	b.	Assuming the time independent Schrodinger wave equation obtain the energy eigen values and eigen function for a Particle in a Box.	L3	CO1	10
10.	a.	Discuss the procedure to find the wavelength of a given LASER experimentally with the necessary equations.	L2	CO2	10
		OR			
	b.	i. Obtain an expression for numerical aperture in terms of refractive indices of core and Cladding and then arrive at the condition for propagation.	L2	CO2	07
		ii. Find the attenuation of light in an optical fiber of length 500m, when a signal of power 100 mW emerges out of the fiber with a power of 90 mW.	L3	CO2	03
11.	a.	i. What is Fermi energy? Explain the Variation of Fermi factor with temperature.	L2	CO3	06
		ii. Calculate the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K and 400 K in a material.	L3	CO3	04
		OR			
	b.	i. With neat diagram, Review the Band theory of Solids.	L2	CO4	05
		ii. Explain the Forward and Reverse bias characteristics of a Zener diode.	L4	CO4	05
12.	a.	Give theory of Damped oscillations and explain the different types of damping with necessary conditions.	L2	CO5	10
		OR			

	b.	Explain the construction and working principle of Reddy`s shock tube and Mention the applications of Reddy`s shock tube.	L3	CO5	**10**
13.	**a.**	Obtain the relation between Young's modulus, rigidity modulus and Poisson's ratio when the force is applied on one face of a cube.	L-3	CO6	**10**
		OR			
	b.	**i.** What are beams? Derive an expression for bending moment of a Beam.	L-3	CO6	**07**
		ii. When a tangential force of 100 N is applied on one of the faces of cube, it displaces by 2 mm. If the side of a cube is 10 cm, calculate the rigidity modulus of a cube.	L-3	CO6	**03**